

Reference Library — Full Summaries

CRLM · Camilo Correa, MD

1999 Clinical score for predicting recurrence after hepatic resection for metastatic colorectal cancer

Fong Y et al. Ann Surg. 1999;230(3):309-318.

QUESTION

How do we quantify recurrence risk before molecular profiling?

P — POPULATION

Patients undergoing hepatic resection for CRC metastases (n=1,001)

I — INTERVENTION

Clinical risk score based on 5 preoperative factors

C — COMPARATOR

N/A - prognostic tool development

O — OUTCOME

Recurrence-free and overall survival stratified by score

KEY RESULTS

5 factors (1 point each): node+ primary, DFI <12 mo, >1 metastasis, CEA >200, size >5 cm. Score 0-2: 5-yr OS ~60%, median OS 74 mo. Score 3-5: 5-yr OS ~14%, median OS 22 mo.

HOW I USE IT

The classic 5-factor clinical risk score. One of the earliest tools to formalize biology-aware surgical selection, before molecular data existed.

<https://pubmed.ncbi.nlm.nih.gov/10493478/>

2004 Rescue surgery for unresectable colorectal liver metastases downstaged by chemotherapy

Adam R et al. Ann Surg. 2004;240(4):644-657.

QUESTION

Can chemotherapy create resectability?

P — POPULATION

Patients with initially unresectable CRLM treated at Paul Brousse

I — INTERVENTION

Downsizing chemotherapy followed by surgical reassessment

C — COMPARATOR

Patients who remained unresectable after chemotherapy

O — OUTCOME

Conversion to resectability rate; long-term overall survival

KEY RESULTS

12.5% conversion rate in patients with initially unresectable disease. 5-yr OS after rescue resection ~33%. Median survival 35 months (resected) vs 16 months (non-resected, $p<0.001$).

HOW I USE IT

Downstaging converts unresectable disease into long-term survivors in selected patients. The conceptual basis for chemotherapy-first in borderline disease.

<https://pubmed.ncbi.nlm.nih.gov/15383792/>

2006 Hepatic arterial infusion versus systemic therapy for hepatic metastases from colorectal cancer (CALGB 9481)

Kemeny NE et al. J Clin Oncol. 2006;24(9):1395-1403.

QUESTION

Does HAI improve outcomes over systemic therapy alone?

P — POPULATION

Patients with unresectable hepatic metastases from CRC (n=135)

I — INTERVENTION

Hepatic arterial infusion chemotherapy (FUDR via implanted pump)

C — COMPARATOR

Systemic chemotherapy (FUDR or 5-FU)

O — OUTCOME

Response rate (primary); overall survival

KEY RESULTS

Response rate 47% (HAI) vs 24% (systemic, $p<0.001$). Median OS 24.4 vs 20.0 months ($p=0.003$). Conversion to resectability 16% vs 4% ($p=0.048$). 2-yr OS 45% vs 32%.

HOW I USE IT

Response rate 47% vs 24% ($p<0.001$); median OS 24.4 vs 20.0 months. One of the few randomized HAI trials showing improved response and conversion in liver-dominant disease.

<https://pubmed.ncbi.nlm.nih.gov/16505413/>

2006 Chemotherapy regimen predicts steatohepatitis and an increase in 90-day mortality after surgery for hepatic colorectal metastases

Vauthey JN et al. J Clin Oncol. 2006;24(13):2065-2072.

QUESTION

Does preoperative chemotherapy cause hepatic injury that increases surgical mortality?

P — POPULATION

Patients undergoing hepatic resection for colorectal metastases (n=406)

I — INTERVENTION

Preoperative chemotherapy (irinotecan-based or oxaliplatin-based)

C — COMPARATOR

No preoperative chemotherapy

O — OUTCOME

Pathologic liver injury (steatohepatitis, sinusoidal dilation); 90-day mortality

KEY RESULTS

Irinotecan: steatohepatitis 20.2% vs 4.4% (OR 5.4, $p<0.001$). Oxaliplatin: sinusoidal dilation 18.9% vs 1.9% (OR 8.3, $p<0.001$). Steatohepatitis: 90-day mortality 14.7% vs 1.6% (OR 10.5, $p=0.001$).

HOW I USE IT

Steatohepatitis from irinotecan-based chemo increases 90-day mortality ~10-fold after hepatectomy. Chemotherapy regimen and duration matter when planning surgery.

<https://pubmed.ncbi.nlm.nih.gov/16648507/>

2006 Complete response of colorectal liver metastases after chemotherapy: does it mean cure?

Benoist S et al. J Clin Oncol. 2006;24(24):3939-3945.

QUESTION

If a liver metastasis disappears on imaging after chemotherapy, is it gone?

P — POPULATION

38 patients where at least one lesion disappeared on CT/ultrasound after chemotherapy

I — INTERVENTION

Surgery with intraoperative ultrasound at sites of imaging complete response

C — COMPARATOR

Sites of disappearing lesions left in place (n=31 LMs)

O — OUTCOME

Residual macroscopic/microscopic disease or in-situ recurrence at sites of imaging CR

KEY RESULTS

83% (55/66) of imaging complete responses had residual disease or in-situ recurrence. Of 31 sites left in place: 23/31 (74%) recurred within 1 year.

HOW I USE IT

Disappearing lesions are not cured lesions. Most will recur if left in situ. Evidence base for resecting all prior disease sites.

<https://pubmed.ncbi.nlm.nih.gov/16921046/>

2007 Actual 10-year survival after resection of colorectal liver metastases defines cure

Tomlinson JS et al. J Clin Oncol. 2007;25(29):4575-4580.

QUESTION

Can surgery actually cure some patients with CRLM?

P — POPULATION

Patients who underwent complete resection of CRLM at a single institution (n=612)

I — INTERVENTION

Hepatic resection with curative intent

C — COMPARATOR

N/A - retrospective cohort defining 10-year cure thresholds

O — OUTCOME

10-year disease-free survival as a proxy for cure

KEY RESULTS

10-year DFS 22.3%. Estimated cure fraction ~17% (plateau of survival curve). Patients disease-free at 5 years had 86% 10-year OS. Median OS 35 months overall.

HOW I USE IT

Roughly 20% of resected patients are disease-free at 10 years. This number is the justification for aggressive surgical approach in CRLM.

<https://pubmed.ncbi.nlm.nih.gov/17925551/>

2008 Perioperative chemotherapy with FOLFOX4 and surgery versus surgery alone for resectable CRLM (EORTC 40983)

Nordlinger B et al. Lancet. 2008;371(9617):1007-1016.

QUESTION

Does perioperative chemotherapy help in resectable CRLM?

P — POPULATION

Patients with resectable CRLM, up to 4 lesions (n=364)

I — INTERVENTION

Perioperative FOLFOX4 (3 cycles pre + 3 post) + surgery

C — COMPARATOR

Surgery alone

O — OUTCOME

Progression-free survival (primary); overall survival

KEY RESULTS

3-yr PFS 36.2% (periop chemo) vs 28.1% (surgery alone), HR 0.77 (p=0.041). Long-term OS 61.3% vs 57.6%, HR 0.88 (p=0.34, NS). Surgical complications higher in chemo arm (25% vs 16%).

HOW I USE IT

PFS improved. OS was not significantly improved at long-term follow-up. Useful for selection and recurrence delay - not a mandate for all patients.

<https://pubmed.ncbi.nlm.nih.gov/18358928/>

2013 EORTC 40983: long-term overall survival results

Nordlinger B et al. Lancet Oncol. 2013;14(12):1208-1215.

QUESTION

Did the PFS signal translate into an OS benefit?

P — POPULATION

Same cohort as EORTC 40983 (2008), long-term follow-up

I — INTERVENTION

Perioperative FOLFOX4 + surgery

C — COMPARATOR

Surgery alone

O — OUTCOME

Overall survival at long-term follow-up

KEY RESULTS

5-yr OS 51.2% (periop chemo) vs 47.8% (surgery alone), HR 0.88 (95% CI 0.68-1.14, p=0.34). OS benefit not demonstrated at long-term follow-up.

HOW I USE IT

Long-term OS remained non-significant. Reinforces careful patient selection rather than automatic perioperative chemotherapy for all resectable patients.

<https://pubmed.ncbi.nlm.nih.gov/24120480/>

2013 Incidence and prognostic impact of KRAS and BRAF mutation in patients undergoing liver surgery for colorectal metastases

Karagkounis G et al. Cancer. 2013;119(23):4137-4144.

QUESTION

What does BRAF V600E do to your enthusiasm for surgery?

P — POPULATION

Patients undergoing liver resection for CRLM with molecular profiling (n=378)

I — INTERVENTION

BRAF V600E or KRAS mutation

C — COMPARATOR

Wild-type BRAF and KRAS

O — OUTCOME

Disease-specific survival, recurrence-free survival

KEY RESULTS

BRAF V600E prevalence 7.4%. Median DSS 24.9 mo (BRAF mut) vs 64.1 mo (WT). 5-yr DSS 24% vs 55% ($p<0.001$). BRAF mutation independently associated with worse DSS on multivariable analysis.

HOW I USE IT

BRAF mutation is uncommon (~5-8% of CRLM) but signals aggressive biology and worse outcomes after resection.

<https://pubmed.ncbi.nlm.nih.gov/24104864/>

2013 Selection for hepatic resection of colorectal liver metastases: expert consensus statement

Adams RB et al. HPB (Oxford). 2013;15(2):91-103.

QUESTION

How should resectability be defined for CRLM in the modern era?

P — POPULATION

Expert consensus panel (AHPBA/SSO/SSAT) reviewing criteria for patient selection and surgical planning in CRLM

I — INTERVENTION

Expert consensus review

C — COMPARATOR

N/A

O — OUTCOME

Consensus criteria for resectability; imaging recommendations; surgical planning principles

KEY RESULTS

Resectability requires adequate liver remnant with perfusion and biliary drainage, and absence of unresectable extrahepatic disease. All sites of prior disease (even imaging CR) should be resected.

HOW I USE IT

The modern operative standard: resectability is defined by adequate remnant and complete eradication, not by lesion count. All prior disease sites should be resected.

<https://pubmed.ncbi.nlm.nih.gov/23297719/>

2014 Systemic chemotherapy with or without cetuximab in patients with resectable colorectal liver metastasis: the New EPOC randomised controlled trial (initial report)

Primrose J et al. Lancet Oncol. 2014;15(6):601-611.

QUESTION

Does adding cetuximab to perioperative chemotherapy improve outcomes in KRAS WT resectable CRLM?

P — POPULATION

Patients with KRAS exon 2 WT resectable or suboptimally resectable CRLM (n=257 analysed)

I — INTERVENTION

Perioperative chemotherapy + cetuximab

C — COMPARATOR

Perioperative chemotherapy alone

O — OUTCOME

Progression-free survival (primary); trial closed early at interim analysis for futility

KEY RESULTS

Median PFS 14.1 months (cetuximab) vs 20.5 months (no cetuximab), HR 1.48 (95% CI 1.04-2.12, p=0.030). Trial stopped early - cetuximab arm worse on the primary endpoint. This is the initial publication that triggered early closure.

HOW I USE IT

The first signal that cetuximab harms in the resectable setting. The 2020 Bridgewater long-term follow-up confirmed OS harm (55 vs 81 months). Together these papers define the resectability boundary as a treatment-effect modifier for anti-EGFR therapy.

<https://pubmed.ncbi.nlm.nih.gov/24717919/>

2015 Effect of KRAS mutation on long-term outcomes after hepatic resection for CRLM

Margonis GA et al. Ann Surg Oncol. 2015;22(13):4158-4165.

QUESTION

How does KRAS change surgical thinking?

P — POPULATION

Patients undergoing hepatic resection for CRLM with known KRAS status (n=1,007, multi-institutional)

I — INTERVENTION

KRAS mutation (any codon)

C — COMPARATOR

KRAS wild-type

O — OUTCOME

Overall survival, recurrence-free survival, recurrence patterns

KEY RESULTS

5-yr OS: KRAS mutant 38.2% vs WT 53.3% ($p<0.001$). 5-yr RFS: 19.3% vs 33.2% ($p<0.001$). Lung recurrence: 37% (mutant) vs 22% (WT, $p<0.001$). Extrahepatic recurrence significantly higher in KRAS mutant.

HOW I USE IT

KRAS mutation is associated with worse survival and more extrahepatic/lung recurrence after hepatectomy. Changes counseling and surveillance - does not prohibit surgery.

<https://pubmed.ncbi.nlm.nih.gov/26077912/>

2015 Phase II trial of hepatic artery infusional and systemic chemotherapy for patients with unresectable hepatic metastases from colorectal cancer: conversion to resection and long-term outcomes

D'Angelica MJ, Correa-Gallego C et al. Ann Surg. 2015;261(2):353-360.

QUESTION

Can HAI + systemic chemotherapy convert unresectable CRLM to resectable, and does conversion improve survival?

P — POPULATION

49 patients with unresectable CRLM (65% previously treated), phase II single-arm trial

I — INTERVENTION

HAI (FUDR) + systemic chemotherapy, with or without bevacizumab

C — COMPARATOR

N/A - single-arm; resected vs non-resected patients compared on landmark analysis

O — OUTCOME

Conversion to resection rate (primary); overall survival, progression-free survival (secondary)

KEY RESULTS

Conversion rate 47% (23/49). Median OS 38 months, median PFS 13 months (all patients). Landmark analysis: 3-yr OS 80% (resected, n=23) vs 26% (non-resected, n=26), $p<0.001$.

HOW I USE IT

47% conversion rate. 3-year OS 80% vs 26% in resected vs non-resected. But this is a selection effect - patients who converted may have had favorable biology independent of surgery. The survival benefit may be in the responders, not the operation itself.

<https://pubmed.ncbi.nlm.nih.gov/24646562/>

2016 Codon 13 KRAS mutation predicts patterns of recurrence after hepatectomy for CRLM

Margonis GA et al. Cancer. 2016;122(17):2698-2707.

QUESTION

Is all KRAS the same?

P — POPULATION

Patients with resected CRLM and known KRAS codon status (n=681)

I — INTERVENTION

KRAS codon 13 mutation specifically

C — COMPARATOR

KRAS codon 12 mutations and KRAS WT

O — OUTCOME

Hepatic, extrahepatic, and pulmonary recurrence patterns

KEY RESULTS

Pulmonary recurrence: codon 13 52% vs codon 12 32% (p=0.02). Codon 13 independently associated with worse RFS (HR 1.68, p=0.02). No significant difference in hepatic-only recurrence between codon 12 and 13.

HOW I USE IT

Codon-specific KRAS biology matters. Codon 13 correlates with worse extrahepatic and lung recurrence - relevant to surveillance planning.

<https://pubmed.ncbi.nlm.nih.gov/27244540/>

2017 Perioperative HAI pump chemotherapy is associated with longer survival after resection of colorectal liver metastases: a propensity score analysis

Groot Koerkamp B et al. J Clin Oncol. 2017;35(17):1938-1944.

QUESTION

Does perioperative HAI improve OS after complete resection of CRLM?

P — POPULATION

Patients with complete resection of CRLM, 1992-2012, single institution (n=2,368)

I — INTERVENTION

Perioperative HAI + systemic chemotherapy (n=785)

C — COMPARATOR

Systemic chemotherapy only (n=1,583); propensity score matched

O — OUTCOME

Overall survival; 10-year OS; subgroup by clinical risk score

KEY RESULTS

Median OS 67 vs 44 months (HAI vs no HAI, propensity matched, HR 0.60, $p < 0.001$). 10-yr OS 38% vs 24%. Effect persisted in the modern systemic chemotherapy era. Strongest benefit: low CRS and node-negative primary.

HOW I USE IT

Median OS 67 vs 44 months with vs without HAI (propensity matched). 10-year OS 38% vs 24%. Effect persisted with modern systemic chemotherapy. Strongest benefit in low clinical risk score and node-negative primary.

<https://pubmed.ncbi.nlm.nih.gov/28426374/>

2019 BRAF mutation is not associated with an increased risk of recurrence after CRLM resection

Bachet J-B et al. Br J Surg. 2019;106(9):1237-1247.

QUESTION

Is BRAF V600E an absolute contraindication to surgery?

P — POPULATION

Patients with BRAF-mutated CRLM who underwent resection (multi-institutional)

I — INTERVENTION

BRAF V600E mutation

C — COMPARATOR

BRAF wild-type patients after resection

O — OUTCOME

Recurrence rate, overall survival, recurrence-free survival

KEY RESULTS

No statistically significant difference in recurrence or OS by BRAF status after R0 resection in this multi-institutional cohort (n=52 BRAF mut). Caveat: small BRAF-mutant sample and retrospective design limit conclusions.

HOW I USE IT

Not necessarily. BRAF should change expectations and counseling - not automatically forbid resection in selected patients.

<https://pubmed.ncbi.nlm.nih.gov/31183866/>

2020 Systemic chemotherapy with or without cetuximab in resectable CRLM (New EPOC)

Bridgewater JA et al. Lancet Oncol. 2020;21(4):e1-e10.

QUESTION

Should cetuximab be added perioperatively in RAS WT resectable CRLM?

P — POPULATION

Patients with resectable RAS WT CRLM (n=272)

I — INTERVENTION

Perioperative chemotherapy + cetuximab

C — COMPARATOR

Perioperative chemotherapy alone

O — OUTCOME

Overall survival; progression-free survival

KEY RESULTS

Median OS 55.4 months (cetuximab) vs 81.0 months (no cetuximab), HR 1.45 (p=0.036). Median PFS 15.5 vs 22.2 months, HR 1.48 (p=0.003). Both endpoints significantly worse with cetuximab.

HOW I USE IT

Cetuximab worsened OS: 55 vs 81 months. A cautionary tale: benefit in unresectable disease does not predict benefit in the resectable setting.

<https://pubmed.ncbi.nlm.nih.gov/32014119/>

2020 Recurrence Patterns After Resection of Colorectal Liver Metastasis are Modified by Perioperative Systemic Chemotherapy

Buisman FE et al. World J Surg. 2020;44(3):876-886.

QUESTION

Does perioperative chemotherapy change recurrence patterns after hepatectomy for CRLM?

P — POPULATION

2020 patients after hepatic resection for CRLM at two centers; 71% received perioperative chemotherapy

I — INTERVENTION

Perioperative systemic chemotherapy

C — COMPARATOR

No perioperative chemotherapy; stratified by clinical risk score (CRS)

O — OUTCOME

Recurrence rate, pattern, and OS by CRS group

KEY RESULTS

High-risk patients (CRS 3-5): fewer extrahepatic recurrences (43% vs 55%, $p=0.007$), especially pulmonary (25% vs 35%, $p=0.007$), and improved OS (HR 0.73, $p=0.02$). No benefit in low-risk patients. No effect on intrahepatic recurrence.

HOW I USE IT

Perioperative chemo reduces pulmonary recurrences and improves OS in high-risk patients (CRS 3-5) only. Supports risk-stratified chemotherapy decision-making.

<https://pubmed.ncbi.nlm.nih.gov/31410513/>

2021 Hepatectomy followed by mFOLFOX6 versus hepatectomy alone for liver-only metastatic CRC (JCOG0603)

Kanemitsu Y et al. J Clin Oncol. 2021;39(34):3789-3799.

QUESTION

Does adjuvant chemotherapy after hepatectomy improve outcomes?

P — POPULATION

Patients after complete resection of liver-only CRC metastases (n=300)

I — INTERVENTION

Hepatectomy + 12 cycles adjuvant mFOLFOX6

C — COMPARATOR

Hepatectomy alone

O — OUTCOME

Disease-free survival (primary); overall survival

KEY RESULTS

Median DFS 49.8 vs 24.9 months, HR 0.56 (95% CI 0.39-0.81, $p < 0.001$). 5-yr OS 83.1% vs 71.2%, HR 0.67 (95% CI 0.45-1.01, $p = 0.073$, NS).

HOW I USE IT

DFS improved markedly (49.8 vs 24.9 months, HR 0.56). OS did not clearly improve. A clear illustration of the DFS-vs-OS debate in surgical oncology.

<https://pubmed.ncbi.nlm.nih.gov/34520230/>

2022 Circulating tumour DNA analysis for improved recurrence surveillance after CRLM resection

Reinert T et al. Int J Cancer. 2022;150(9):1537-1548.

QUESTION

Does ctDNA predict recurrence earlier than imaging?

P — POPULATION

Patients after curative resection of CRLM with ctDNA monitoring

I — INTERVENTION

Postoperative tumor-informed ctDNA surveillance

C — COMPARATOR

Standard imaging-based surveillance

O — OUTCOME

Lead time to recurrence detection; prognostic accuracy

KEY RESULTS

ctDNA-positive post-resection: 3-yr RFS 0% vs 44% ctDNA-negative (HR 10.0, $p < 0.001$). Lead time advantage 6.1 months before imaging detection of recurrence. Sensitivity 91%, specificity 88%.

HOW I USE IT

ctDNA detects recurrence 6-12 months before imaging in many patients and outperforms standard clinical risk factors as a prognostic tool.

<https://pubmed.ncbi.nlm.nih.gov/34994972/>

2022 Clinicopathological and prognostic significance of HER2 status in surgically resected colorectal liver metastases

Han J et al. J Surg Oncol. 2022;125(6):991-1001.

QUESTION

Does HER2 status affect prognosis after hepatectomy for CRLM?

P — POPULATION

Patients with surgically resected CRLM with known HER2 status by IHC and FISH (n=208)

I — INTERVENTION

HER2 amplification by FISH (IHC 2+/3+ confirmed)

C — COMPARATOR

HER2 non-amplified CRLM

O — OUTCOME

Relapse-free survival; overall survival; association with clinicopathological features

KEY RESULTS

HER2 amplification prevalence 6.25%. In left-sided RAS/RAF WT patients: worse OS (30.2 vs 50.9 months, $p=0.040$) and RFS (5.77 vs 19.97 months, $p=0.017$). Associated with younger age, bilobar involvement, left-sided RAS/RAF WT.

HOW I USE IT

HER2 is relevant in ~6% of resected CRLM. In left-sided RAS/RAF WT patients it predicts worse survival after resection and is an actionable target.

<https://pubmed.ncbi.nlm.nih.gov/35150441/>

2022 Genomic Stratification of Resectable Colorectal Liver Metastasis Patients and Implications for Adjuvant Therapy and Survival

Ecker BL et al. Ann Surg. 2022;275(2):371-381.

QUESTION

Can genomic risk stratification guide selection for adjuvant HAI vs systemic chemotherapy alone after resection of CRLM?

P — POPULATION

334 patients with resected CRLM and somatic mutation profiling (MSK-IMPACT); SYS+HAI-FUDR (n=204) vs SYS alone (n=130)

I — INTERVENTION

Adjuvant SYS+HAI-FUDR (hepatic artery infusion of floxuridine + systemic chemotherapy)

C — COMPARATOR

Adjuvant systemic chemotherapy alone

O — OUTCOME

Overall survival by genomic risk group (low/moderate/high based on RAS/RAF, SMAD4, EGFR alterations)

KEY RESULTS

Adjuvant HAI improved OS only in the low-risk genomic group (HR 0.50, p=0.045); propensity-matched 5-yr OS 89% vs 68% (p=0.019). No benefit in moderate- or high-risk groups. HAI benefit is genomically restricted.

HOW I USE IT

Genomic stratification predicts who benefits from adjuvant HAI. Low-risk patients (RAS/RAF WT, SMAD4 intact) derive the greatest survival benefit. High-risk mutations may negate the HAI advantage.

<https://pubmed.ncbi.nlm.nih.gov/34793355/>

2023 Tucatinib plus trastuzumab for chemotherapy-refractory, HER2-positive, RAS wild-type unresectable or metastatic colorectal cancer (MOUNTAINEER)

Strickler JH et al. Lancet Oncol. 2023;24(5):496-508.

QUESTION

Does tucatinib + trastuzumab have activity in HER2+ RAS WT mCRC?

P — POPULATION

Patients with chemotherapy-refractory HER2-positive, RAS WT unresectable or metastatic CRC (n=84 evaluable in cohorts A+B)

I — INTERVENTION

Tucatinib + trastuzumab

C — COMPARATOR

Tucatinib monotherapy

O — OUTCOME

Confirmed objective response rate by BICR; OS; PFS

KEY RESULTS

ORR 38.1% (95% CI 27.7-49.3%) in cohorts A+B. FDA-approved first anti-HER2 regimen for mCRC. Durable responses in a previously refractory population.

HOW I USE IT

First FDA-approved HER2-directed therapy for mCRC. Relevant for patients with HER2-amplified CRLM who progress after resection.

<https://pubmed.ncbi.nlm.nih.gov/37142372/>

2024 Personalized circulating tumor DNA monitoring improves recurrence surveillance after curative resection of CRLM

Li Y et al. Int J Surg. 2024.

QUESTION

Can ctDNA change surveillance and salvage strategy?

P — POPULATION

Patients after curative resection of CRLM (prospective cohort)

I — INTERVENTION

Personalized ctDNA monitoring protocol

C — COMPARATOR

Conventional surveillance (imaging + CEA)

O — OUTCOME

Recurrence detection sensitivity and specificity; impact on salvage therapy

KEY RESULTS

ctDNA sensitivity 82%, specificity 94% for recurrence detection. PPV 91%. Median lead time 5.5 months before clinical recurrence. ctDNA-guided strategy changed clinical management in 38% of cases.

HOW I USE IT

Prospective data support ctDNA for early recurrence detection and may guide decisions about repeat local therapy.

<https://pubmed.ncbi.nlm.nih.gov/38445460/>

2024 Long-term follow-up from the randomized LIGRO trial (ALPPS vs two-stage hepatectomy)

Sandstrom P et al. Ann Surg. 2024.

QUESTION

Can ALPPS create resectability with meaningful long-term outcomes?

P — POPULATION

Patients with bilobar CRLM requiring staged resection (LIGRO RCT)

I — INTERVENTION

ALPPS (associating liver partition and portal vein ligation for staged hepatectomy)

C — COMPARATOR

Two-stage hepatectomy with portal vein occlusion

O — OUTCOME

Resectability rate (primary); overall survival at long-term follow-up

KEY RESULTS

Resectability rate 92% (ALPPS) vs 57% (TSH), $p < 0.001$. 5-yr OS comparable (31% vs 29%, NS). 90-day major morbidity higher with ALPPS (43% vs 26%). Median OS not significantly different.

HOW I USE IT

ALPPS achieves faster hypertrophy and higher R0 resection rates than two-stage hepatectomy, with comparable long-term outcomes in selected patients.

<https://pubmed.ncbi.nlm.nih.gov/39310365/>

2025 Thermal ablation versus surgical resection of small-size CRLM (COLLISION trial)

COLLISION Trial Group. Lancet Oncol. 2025.

QUESTION

Is resection always superior for small resectable lesions?

P — POPULATION

Patients with resectable CRLM up to 3 cm (randomized, multi-center)

I — INTERVENTION

Thermal ablation (microwave or radiofrequency)

C — COMPARATOR

Surgical resection

O — OUTCOME

Overall survival (non-inferiority primary); local recurrence rate

KEY RESULTS

3-yr OS 80.6% (ablation) vs 80.0% (resection), non-inferiority margin 10% met. Local recurrence 11.5% (ablation) vs 4.0% (resection, $p < 0.001$). No significant difference in distant recurrence or DFS.

HOW I USE IT

For lesions up to 3 cm, thermal ablation is non-inferior to resection for OS. Supports parenchymal-sparing ablation for small peripheral lesions in selected patients.

<https://pubmed.ncbi.nlm.nih.gov/39848272/>

2026 Critical Evaluation of Treatment Response, Driver Mutations, and Circulating Tumor DNA as Markers of Tumor Biology in Colorectal Liver Metastasis

Madi M et al. Cancers (Basel). 2026;18(7):1111.

QUESTION

How should treatment response, driver mutations, and ctDNA be integrated into CLM management?

P — POPULATION

Systematic review of the literature on tumor biology markers in CLM (Medline, Embase, Cochrane)

I — INTERVENTION

Review of chemotherapy response (radiologic/pathologic), driver mutations (RAS/BRAF/TP53/SMAD4/FBXW7/APC), and ctDNA

C — COMPARATOR

N/A - narrative/systematic review

O — OUTCOME

Association of each marker with OS, RFS, and clinical decision-making

KEY RESULTS

Six driver mutations consistently affect outcomes. A three-tier pathway-centric risk score integrating RAS/BRAF, TP53, and SMAD4/FBXW7 offers the most comprehensive approach. Preoperative ctDNA predicts worse survival; clearance after hepatectomy correlates with improved OS. Mutations should not influence margin strategy or parenchymal sparing.

HOW I USE IT

A current framework for integrating biology markers into CLM decision-making: response, mutations, and ctDNA each provide complementary information. Supports biology-first thinking over anatomy-first.

<https://pubmed.ncbi.nlm.nih.gov/41976334/>